Exp. No:

Capacitance of a Parallel Plate Capacitor in Different Dielectric Media Date

Aim:

To write program for capacitance of parallel plate capacitor in three different dielectric medium with specific area, dielectric thickness and various dielectric constants using MATLAB.

Objectives:

1. To compute the capacitance of a parallel plate capacitor for varying plate area, separation and dielectric constant.

2. To compare the capacitance obtained across three different dielectric mediums.

3. To validate the simulated results using Poisson's and Laplace's equation based theoretical values.

Test case (i) Determination and plotting of capacitance of parallel plate capacitor for given area and dielectric thickness

Program:

```
clc;
clear;
A=input("Enter the area value=");
d=input("Enter the thickness value=");
epsilon0=8.854*10^-12;
C=epsilon0*A/d;
disp('The capacitance is');
disp(C);
% Plot: capacitance versus plate separation for the given area
dv = linspace(1e-3,10e-3,50);
Cv = epsilon0*A./dv;
```

figure;

plot(dv*1e3,Cv,'LineWidth',2);

grid on;

xlabel('Plate separation d (mm)'); ylabel('Capacitance (F)');

title('Parallel plate capacitance versus separation');

% Validation using Laplace's equation (1-D finite-difference solution between the plates)

V0 = 100; N = 51; V = zeros(1,N); V(N) = V0;

for it = 1:20000

V(2:N-1) = 0.5*(V(1:N-2) + V(3:N));

% Laplace equation: d2V/dx2 = 0 end x = linspace(0,d,N);

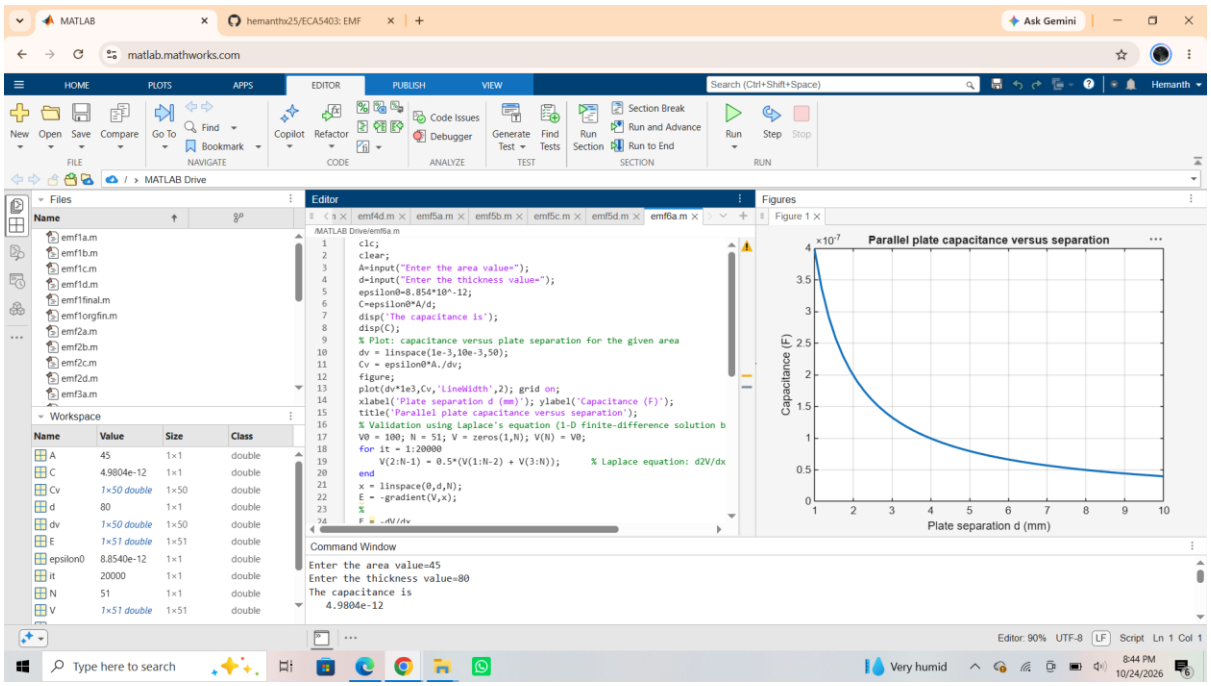
E = -gradient(V,x);

% E = -dV/dx C_lap = epsilon0*A*abs(mean(E))/V0;

% C = Q/V0 with Q = epsilon0*E*A

fprintf('Capacitance from Laplace solution = %.4e F\n', C_lap);

OUTPUT:

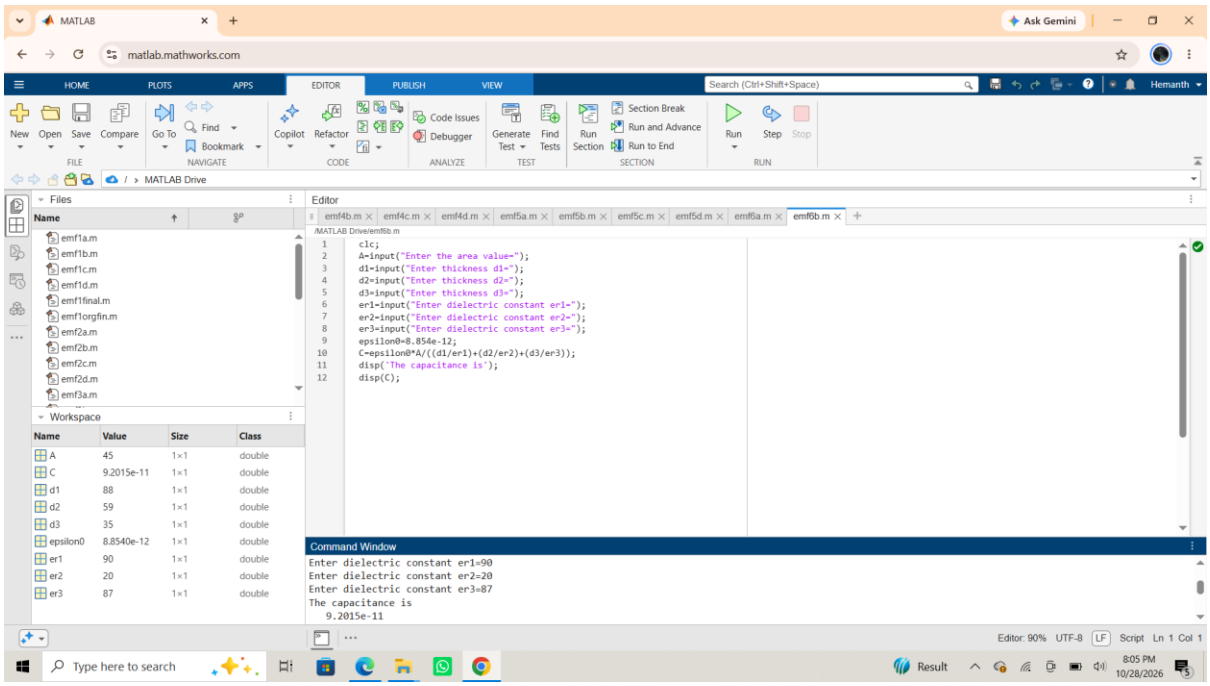


Test case (ii) Determination of capacitance of parallel plate capacitor with three different dielectric layers

Program:

```
clc;
A=input("Enter the area value=");
d1=input("Enter thickness d1=");
d2=input("Enter thickness d2=");
d3=input("Enter thickness d3=");
er1=input("Enter dielectric constant er1=");
er2=input("Enter dielectric constant er2=");
er3=input("Enter dielectric constant er3=");
epsilon0=8.854e-12;
C=epsilon0*A/((d1/er1)+(d2/er2)+(d3/er3));
disp('The capacitance is');
disp(C);
```

OUTPUT:

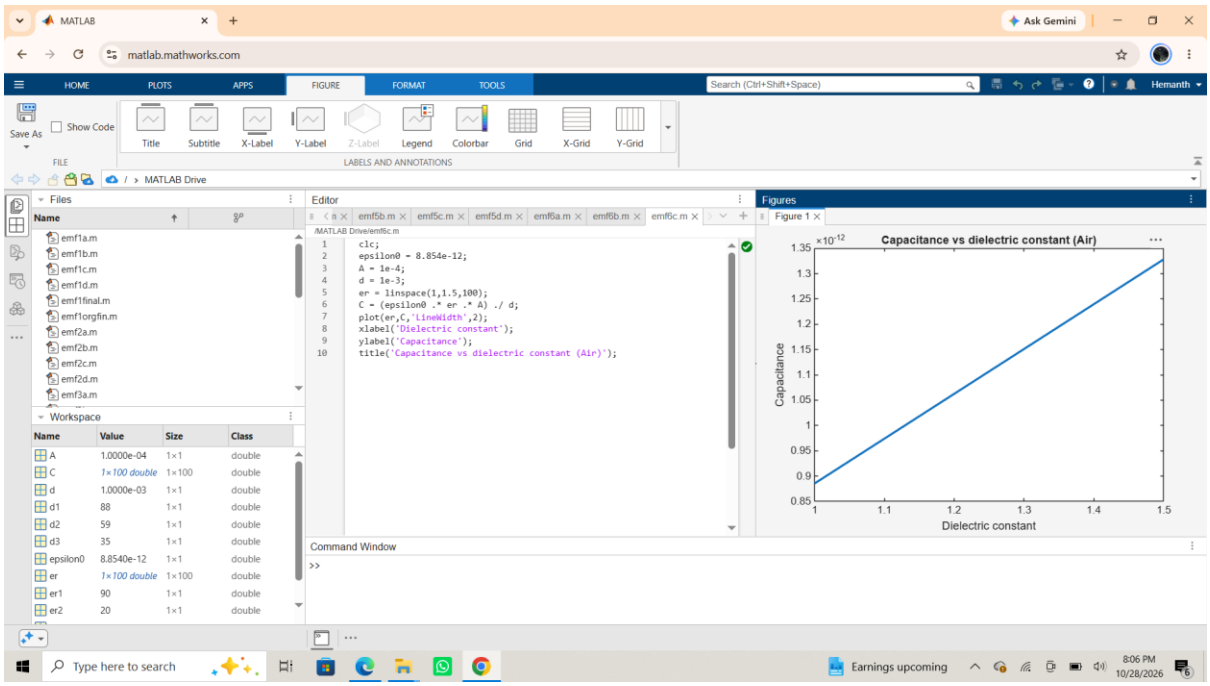


Test case (iii) Determination and plotting of capacitance variation with dielectric constant (Air)

Program:

```
clc;
epsilon0 = 8.854e-12;
A = 1e-4;
d = 1e-3;
er = linspace(1,1.5,100);
C = (epsilon0 .* er .* A) ./ d; plot(er,C,'LineWidth',2);
xlabel('Dielectric constant');
ylabel('Capacitance');
title('Capacitance vs dielectric constant (Air)');
```

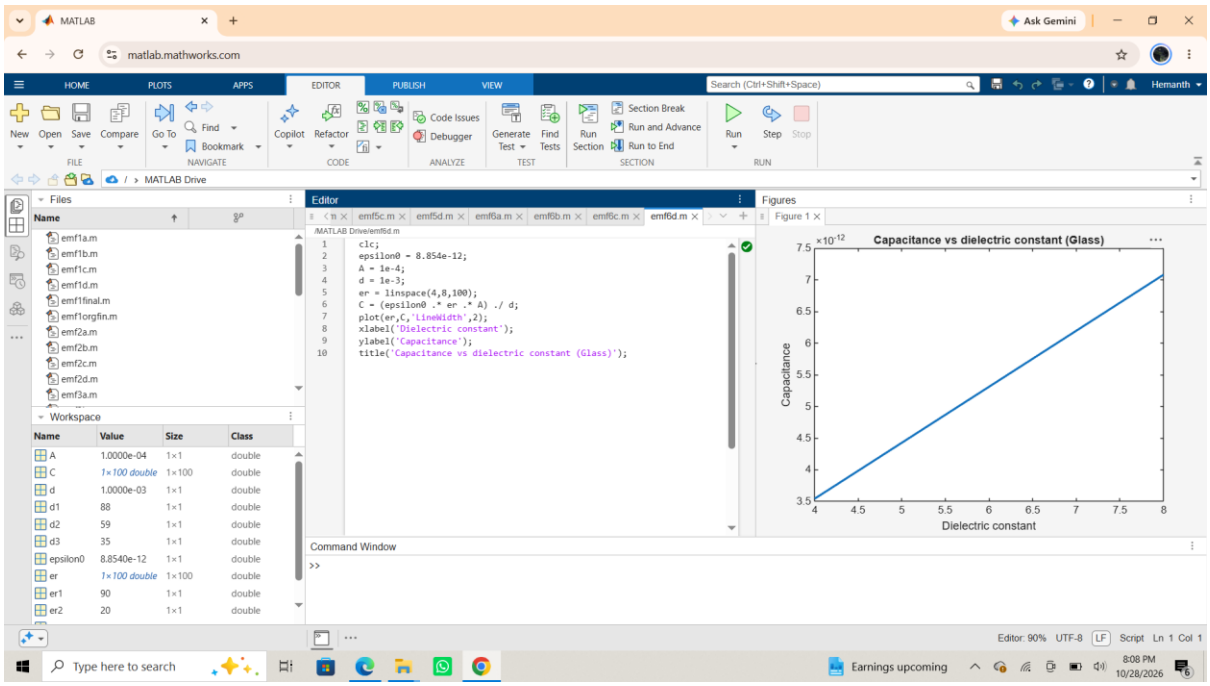
OUTPUT:



Test case (iv) Determination and plotting of capacitance variation with dielectric constant (Glass)

```
clc;
 epsilon0 = 8.854e-12;
A = 1e-4;
 d = 1e-3;
er = linspace(4,8,100);
 C = (epsilon0 .* er .* A) ./ d; plot(er,C,'LineWidth',2);
 xlabel('Dielectric constant');
 ylabel('Capacitance');
 title('Capacitance vs dielectric constant (Glass)');
```

OUTPUT:



S. No.	Objective	Assessment Criterion	Excellent – 4	Good – 3	Satisfactory – 2	Needs Improvement – 1	Weighting (%)	Marks given	Marks obtained

1	Set capacitor geometry and material properties	Set capacitor geometry and material properties	. Correctly enters plate area, separation and relative permittivity with SI units.	. Inputs are correct with a minor unit/pr o perty issue.	Several inputs need correction.	Geometry or dielectric inputs are invalid.	25	2.5	
2	Calculate capacitan c e in MATLA B	Calculate capacitanc e in MATLAB	Uses the correct capacitance relation and calculates values for all specified media.	. Calcula t ions are mostly correct with minor errors.	. Only some cases are calculated correctly.	Capacitance calculations are incorrect.	25	2.5	
3	Compare dielectric and geometry effects	Compare dielectric and geometry effects	Correctly evaluates trends versus permittivity/are a/ separation and validates the model.	Trend and validati on are mostly correct with minor omissi o ns.	Trend discussion or validation is incomplete .	. Incorrectly interprets parameter effects.	30	3	
4	Docume nt engineeri n g observati o ns	Document engineerin g observatio n s	. Provides a complete table/plot with units and explains dielectric influence.	Docum e ntation and explan at ion are mostly comple t e.	. Some labels, units or explanatio ns are missing	Some labels, units or explanations are missing	20	2	
Tota l							100	10	